ON THE ELECTRONIC STRUCTURE OF EXCITED NEUTRAL ARGON CLUSTERS

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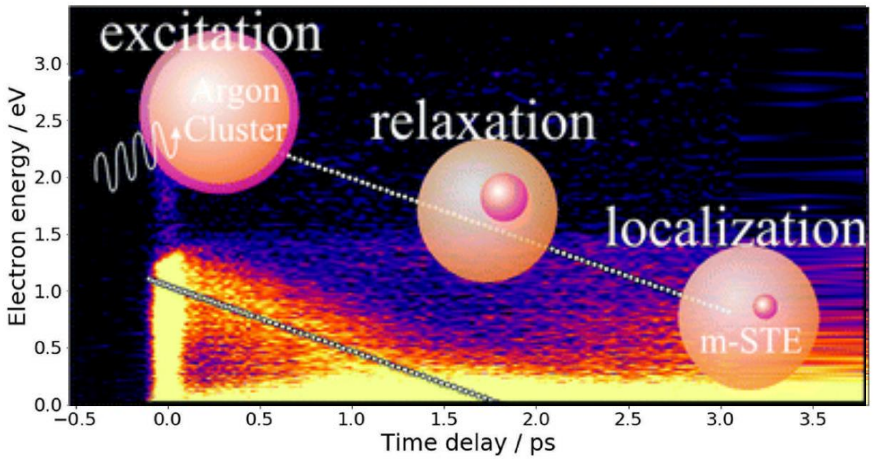


SIMUL GROUP:

- THEORETICAL GROUP WORKING WITH EXPERIMENTAL GROUPS
- MODELING AND SIMULATIONS WITH IN HOUSE CODES
- ELECTRONIC STRUCTURE AND MOLECULAR DYNAMICS
- ION BEAM TRANSPORT IN CAPILLARIES
- DI-CATION DISSOCIATION IN STRONG ELECTRIC FIELD

MOTIVATION

- To follow the relaxation path of excitation created in large (N∼150 atoms) neutral Argon cluster [1]
  - Huge configuration space for full dynamics
  - Singlet and triplet states are coupled
  - Need approximations
  - *Ab initio* for one electron in multiple orbitals is not easy
  - To do dynamics we need to obtain lowest energy structures
  - Methods to get on-the-fly reliable potential curves
- Why we look for lowest energy structure?
  - Previous work by Naumkin [2] has reported lowest energy structures
  - **We tried to do the same more carefully (using same method and almost *ab initio* method) and we observe they are not same and here we want to report that.**



[1] A. Lietard et al., doi: 10.1039/c7cp06789e
[2] F. Y. Naumkin and D. J. Wales, doi: 10.1080/00268979909483074.

THE MODEL

- The excitation in formed by promoting one valance electron an Argon to some excited state, in our case $3p^5 4s$ state
- Excimer cluster can be treated as ionic Argon cluster with an added Rydberg electron.
- The lowest molecular states of the diatomic states are ${}^{1,3}\Sigma_{u,g}^{+}$ and ${}^{1,3}\Pi_{u,g}$ of which triplet states are observed to be lowest.
- We are going to neglect spin-orbit coupling; hence singlet and triplet states can be separated.

ON-THE-FLY POTENTIAL ENERGY CURVES

- Why and what method to use it get on-the-fly PEC's:
  - To get a minimum structures of higher number of atoms, the use of some ab initio method is computationally expensive, hence we relay on some approximate method.

  - Here we are going to use Diatomic-In-Molecules (DIM) method:
    - The electronic energies of n-atomic system is based on previously calculated atomic and diatomic fragments
    - Very fast
    - For ionic clusters shown to be equivalent to *ab initio* [3]
    - It is a "semi-empirical" method based on valance bond theory (VBT)
    - significantly reduced configuration space of VBT ($\sim 10^2$ configs)
    - This method solves the non-relativistic time-independent Schrodinger equation for electrons in fixed potential of Nuclei i.e., using Bohn-Oppenheimer approximation.

[3] Naumkin, F. Y., Knowles, P. J., & Murrell, J. N. (1995). https://doi.org/10.1016/0301-0104(94)00414-6

DIM METHOD

- The idea of DIM was proposed by Ellison[4] in which a polyatomic molecule potential energy surface can be estimated by selecting suitable atomic and diatomic potential energy surfaces
- A valance bond like polyatomic basis functions by having a direct anti-symmetrized product of spin and symmetry adapted atomic basis functions

$$\Phi_m = \mathcal{A}_n \varphi(1,2,\dots n) = \mathcal{A}_n A_{\alpha i} B_{\beta j} C_{\gamma k} \dots$$

- The block Hamiltonian to form atomic and diatomic fragments

$$\widehat{H} = \sum_{B}^{N} \sum_{A>B}^{N} H_{AB} - (N-2) \sum_{B}^{N} H_B$$

- Parametrization:
  - The atomic and diatomic fragments can be obtained from some *ab initio* methods (CCSDT, MRCI) using some quantum chemistry codes.
  - For our system we get from: Hole-Particle core Pseudo-Potential method

[4] F. O. Ellison, doi: 10.1021/ja00905a002

PARAMETRIZATION AND VALIDATION OF DIM: HOLE-PARTICLE PSEUDOPOTENTIAL (HPP) METHOD

Proposed by P. Duplàa and F. Spiegelman [5]

Key features:

The basic idea is to present a hole-particle Hamiltonian for rare gas excimers with single excitation.

$$|\Omega_{A\mu\zeta}^{i\alpha}\rangle = a_{i\alpha}^{\dagger} a_{A\mu\zeta} |\Omega_0\rangle$$

$$H_{A\mu\zeta,B\mu\sigma}^{i\alpha,j\beta} = \left\langle \Omega_{A\mu\zeta}^{i\alpha} \middle| H_{el} \middle| \Omega_{B\mu\sigma}^{j\beta} \right\rangle$$

A delocalized Rydberg orbital is added which serves as an excited state.

The holes are treated as in ionic DIM model where they are supposed to be localized on the atomic sites, but no assumption like this is taken for particle i.e. Rydberg electron.

The Hamiltonian created is two-fold in which hole contributions are treated as DIM model and excited electron as one-electron pseudopotential where all the electrons in valance band are considered in core except for the Rydberg electron.

The Hamiltonian considers following types of coupling, core-core interactions, core-Rydberg interactions, hole transfer and excitation transfer.

[5] P. Duplàa and F. Spiegelmann, doi: 10.1063/1.472012

PARAMETRIZATION AND VALIDATION OF DIM: HOLE-PARTICLE PSEUDOPOTENTIAL (HPP) METHOD

Key features (continued):

The two-electron contributions, atomic intra-multiplets splitting, intra and inter-configuration molecular couplings are considered in simplified scheme.

Core-polarization pseudopotential (CPP) are used for core-polarization and core-Rydberg correlations.

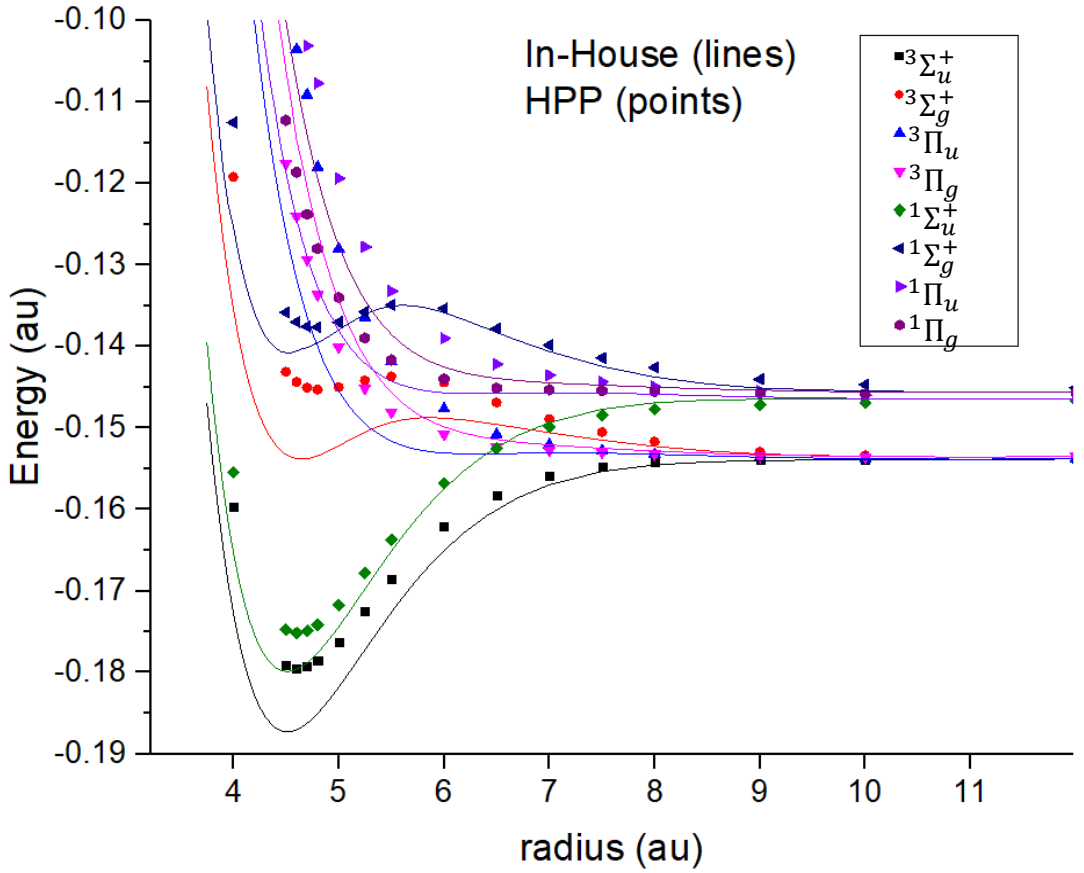
Due to the diffused Rydberg orbital in any number of higher states, the selection of states in which it can reside changes the potential energy curves. Hence, higher the better.

In-House HPP Model:

This is developed to get better potential energy curves to feed the DIM model for large neutral excited Argon clusters.

In our model we consider much more diffused Rydberg states adiabatically disassociated up to 3d configurations in whole distance range by having larger basis set.

The ionic potential curves for DIM orbitals are obtained using CCSDT ab initio calculations using quantum chemistry program, MOLCAS, while considering basis-set superposition error.



[5] P. Duplàa and F. Spiegelmann, doi: 10.1063/1.472012

CURING POTENTIAL ENERGY CURVES

The Potential Curves obtained from ab-initio methods are adiabatic, i.e. they do not represent pure atomic states and are coupled with the higher states

An avoided crossing between states ${}^{1,3}\Sigma_g$ and ${}^{1,3}S_g$ is observed, hence they are not diabatic potential curves which are important for accuracy of DIM.

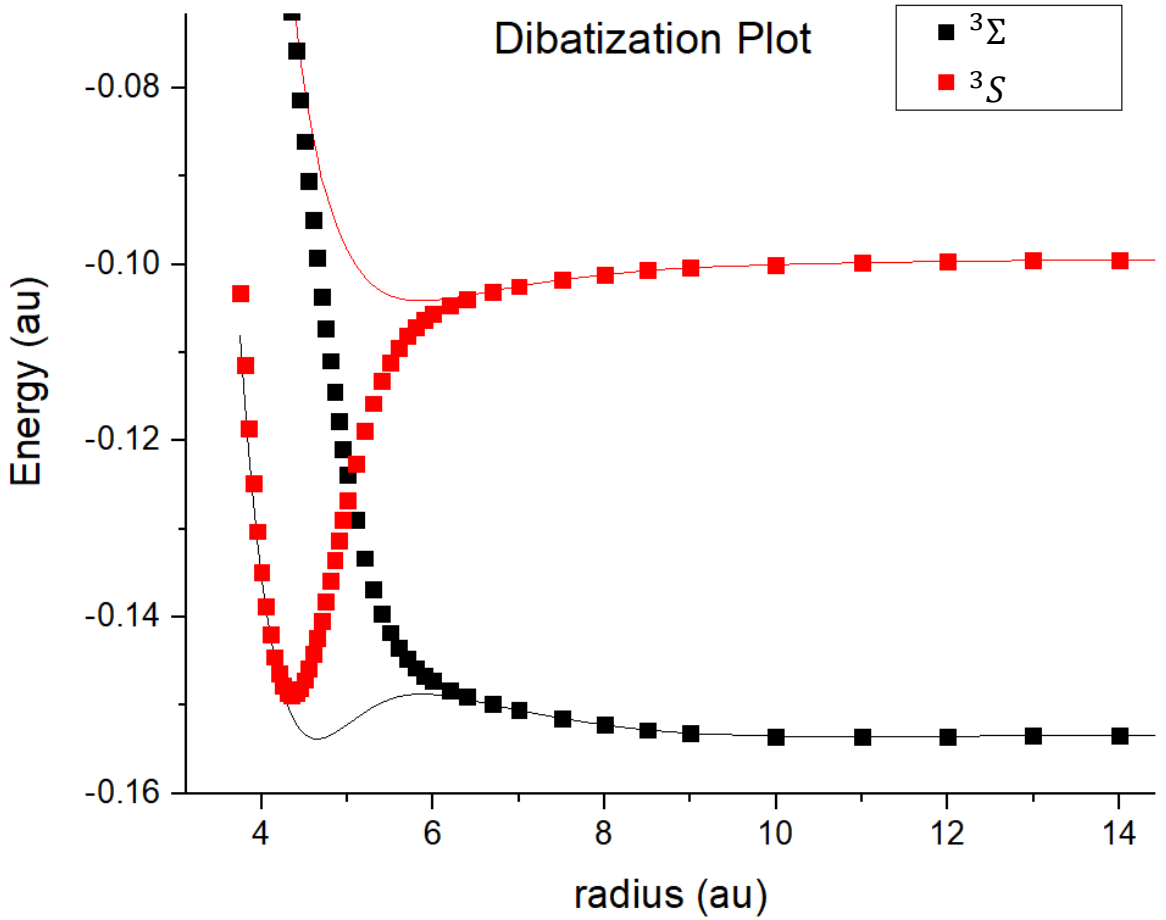
Using direct diabatization method to change the adiabatic V_{ad} to diabatic potential V_d as,

$$V_d = U^{\dagger} V_{ad} U$$

where U is the rotation matrix

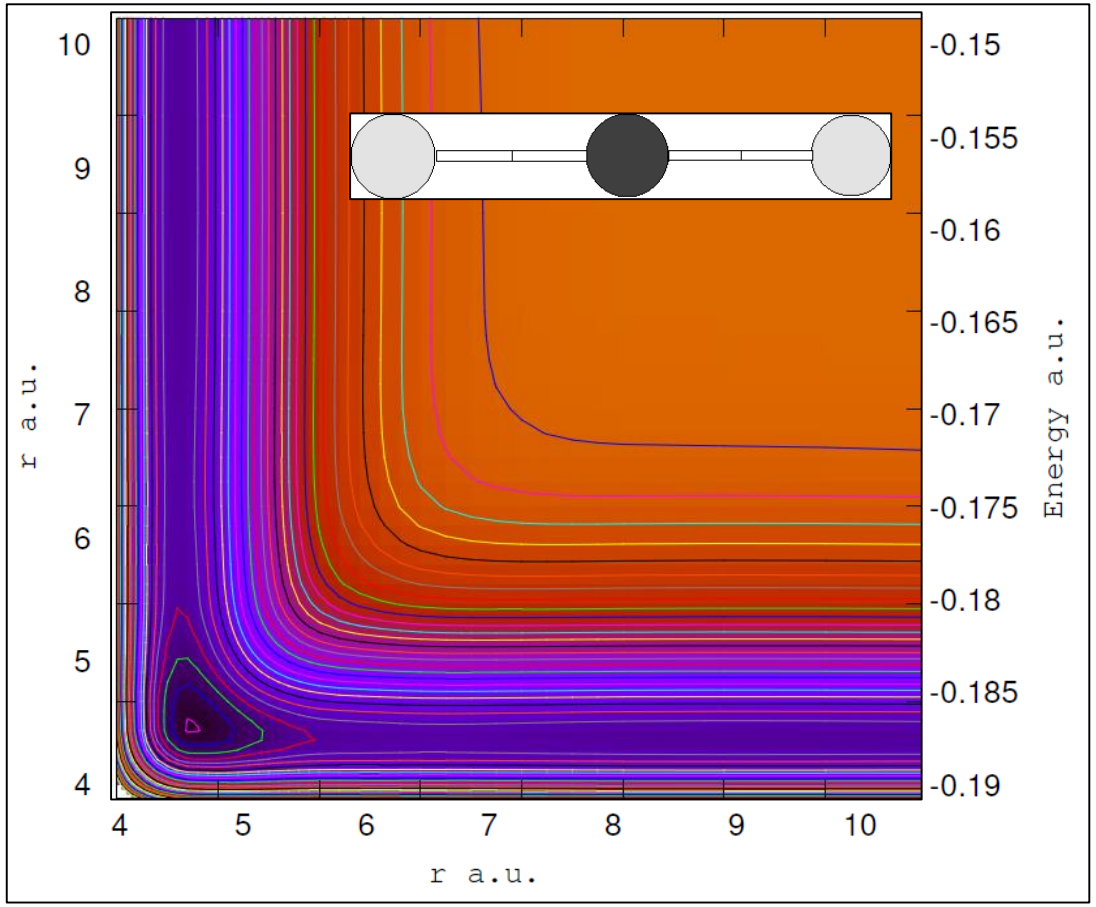
We don't know the real coupling between the higher potentials and hence we supply coupling using ad-hoc method and eye-balling best diabatic curves that should follow the characteristics of the coupled states

The validation of supplied couplings is done by comparing the results of Trimers with HPP method.

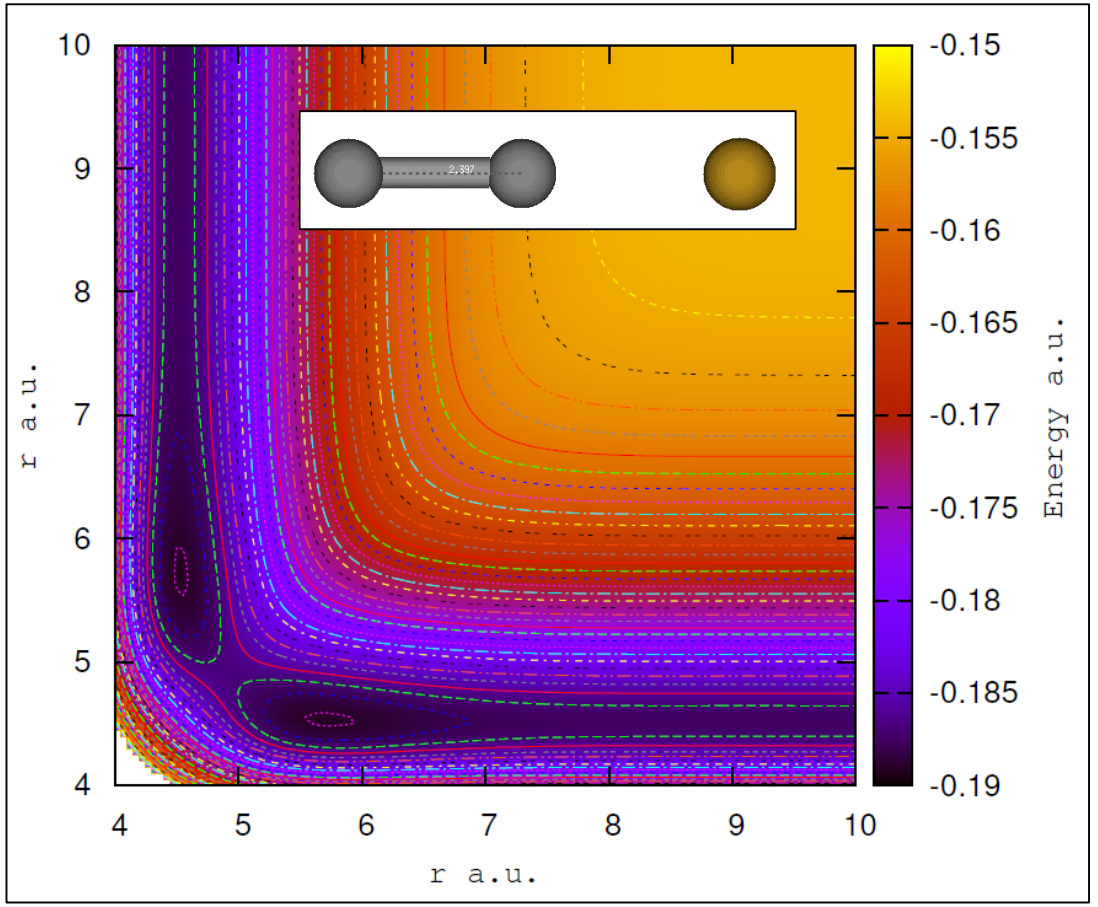


SIGNIFICANCE OF DIABATIZATION

Without Diabatization Naumkin's [2] results



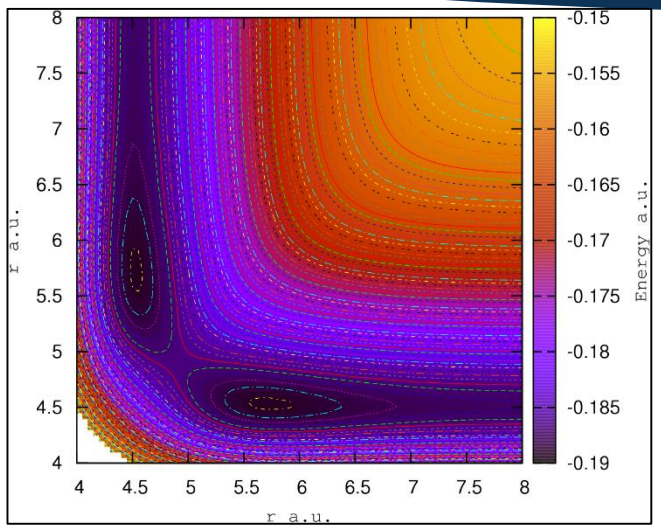
With Diabatization

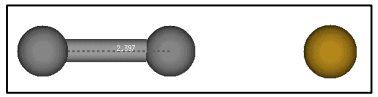


[2] F. Y. Naumkin and D. J. Wales, doi: 10.1080/00268979909483074

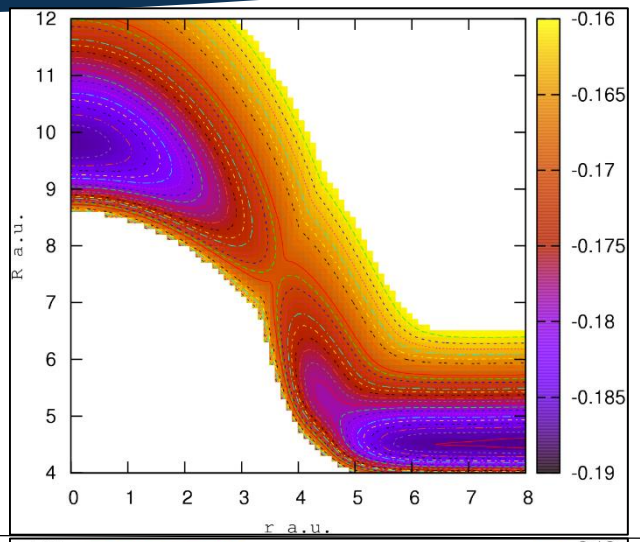
VALIDATION OF DIM MODEL AND DIABATIZATION LINEAR ANTI SYMMETRIC

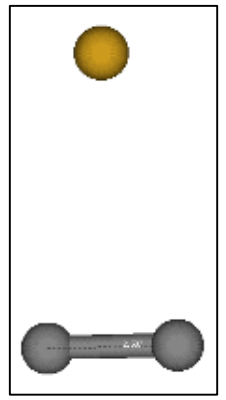
DIM Model





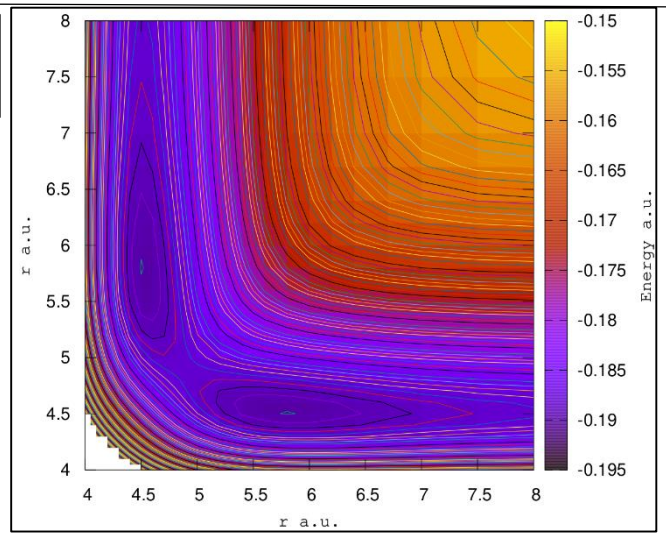
Linear Anti-Symmetric

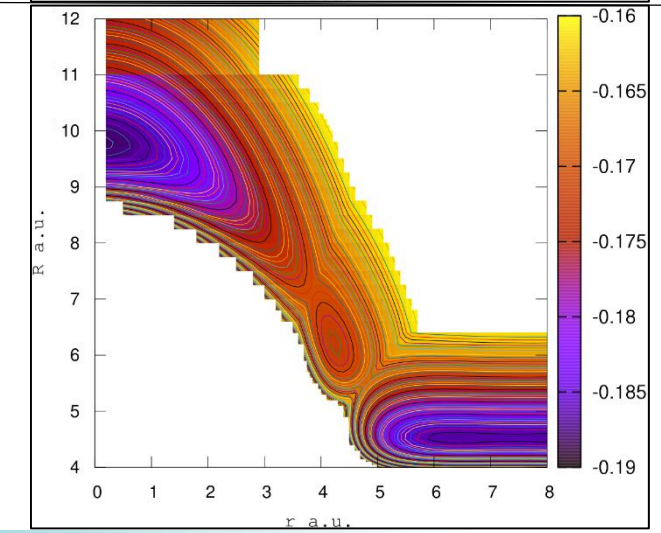




Triangular C2v

HPP Model

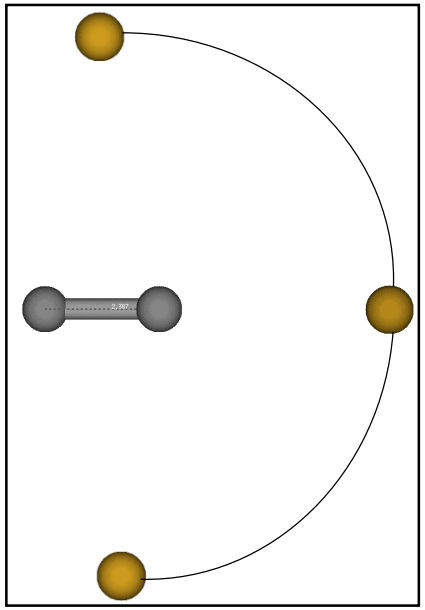


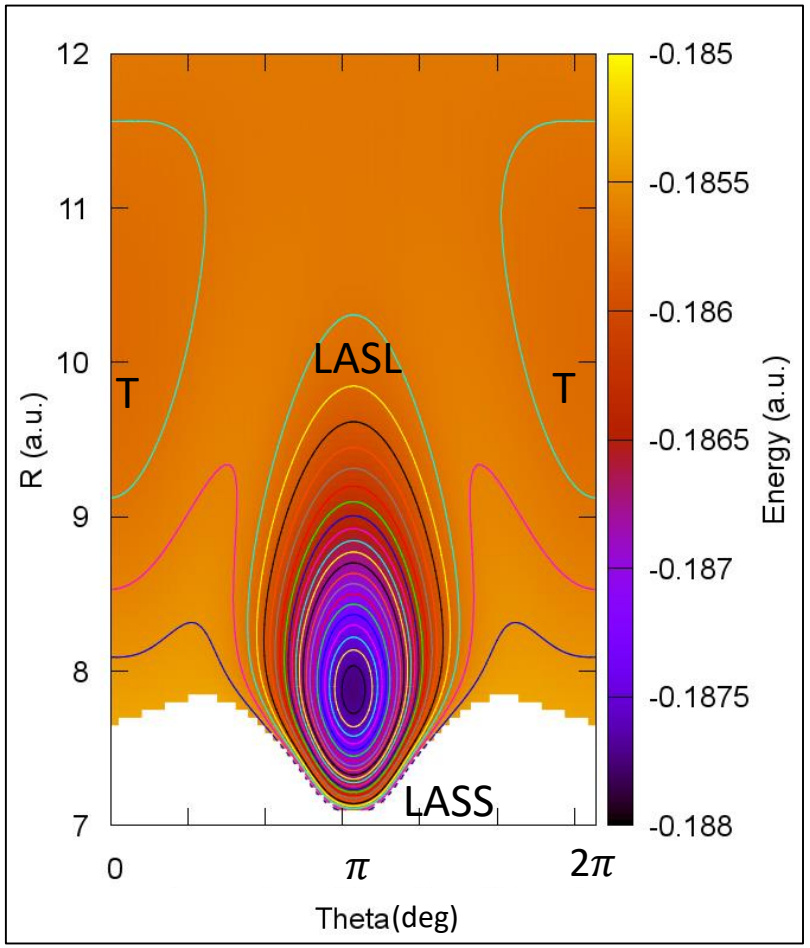


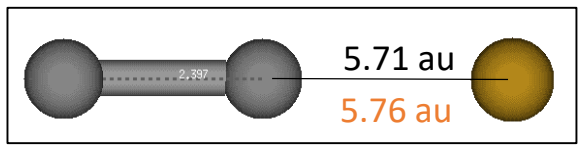
MORE ABOUT TRIMERS

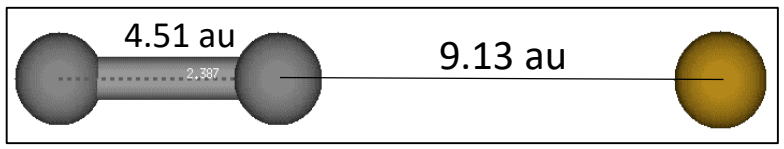
Isomer HPP	Energy (au)	Energy diff w.r.t lowest isomer	Excimer	Order
Anti-Symmetric short	-0.19007293		Surface	1
Triangular	-0.18754384	0.00252909	Surface	2
Anti-Symmetric Long	-0.18783613	0.0022368	Surface	3

Isomer DIM	Energy (au)	Energy diff w.r.t lowest isomer	Excimer	Order
Anti-Symmetric short	-0.189112709		Surface	1
Triangular	-0.187637343	0.001475366	Surface	2
Anti-Symmetric Long	-0.187563881	0.001548829	Surface	3





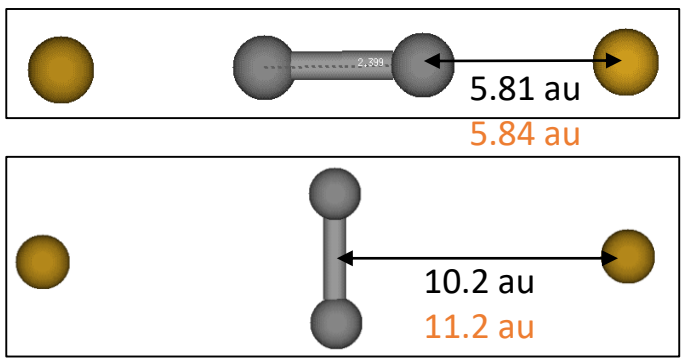


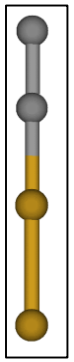


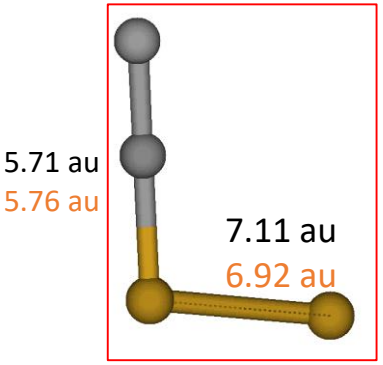
ISOMERS FOR TETRAMERS

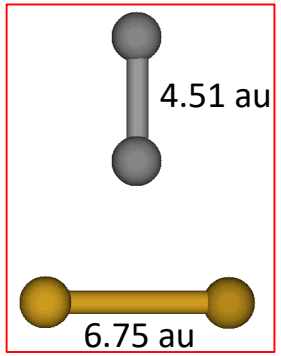
Isomer HPP	Energy (au)	Energy diff wrt lowest isomer	Excimer	Order HPP
linear Symmetric	-0.19245869		Core	1
T	-0.19142072	0.00103797	Surface	2
linear anti symmetric	-0.19083575	0.00162294	Surface	3
Naumkin "L"	-0.19079604	0.00166265	Surface	4
3D shape	-0.18817536	0.00428333	Surface	5
planar	-0.18816019	0.0042985	Surface	6
losange	-0.18773355	0.00472514	Core	7

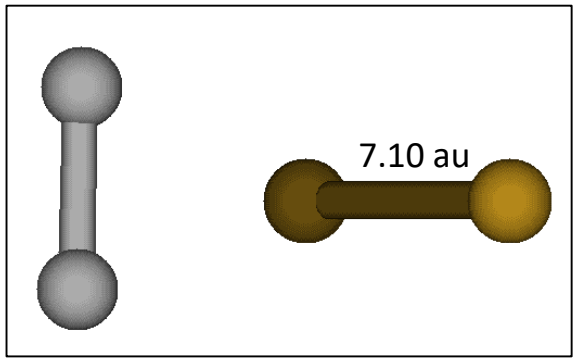
Isomer DIM	Energy (au)	Energy diff wrt lowest isomer	Excimer	Order HPP
linear Symmetric	-0.190623006	0	Core	1
L – Shape (Naumkin)	-0.189812878	0.000810128	Surface	4
linear anti symmetric	-0.189698318	0.000924687	Surface	3
T	-0.189306599	0.001316407	Surface	2
3D shape	-0.188431231	0.002191775	Surface	5
Planer	-0.188382027	0.002240978	Surface	6
losange	-0.187979906	0.002643099	Core	7





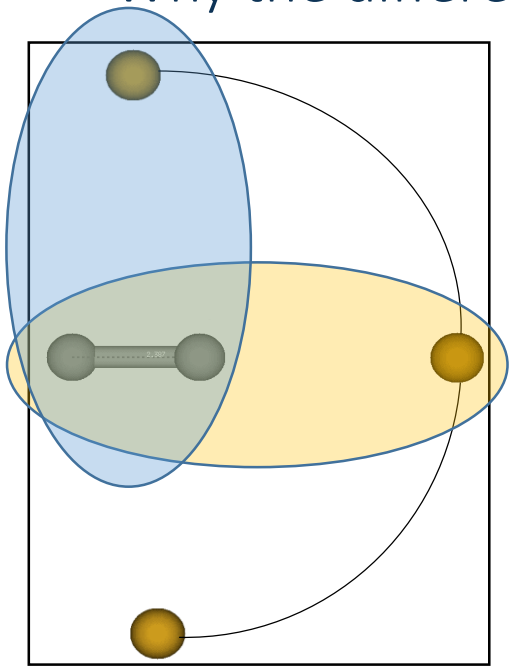


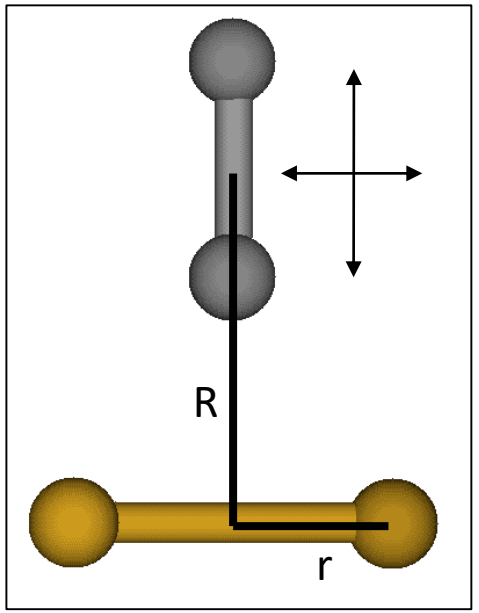




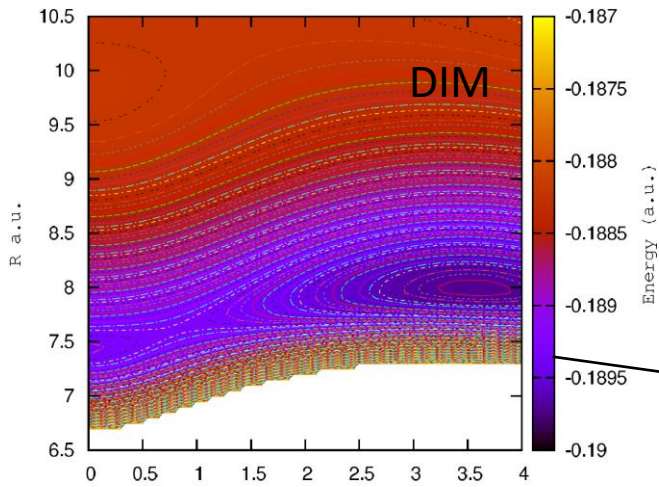
TETRAMERS HPP VS DIM

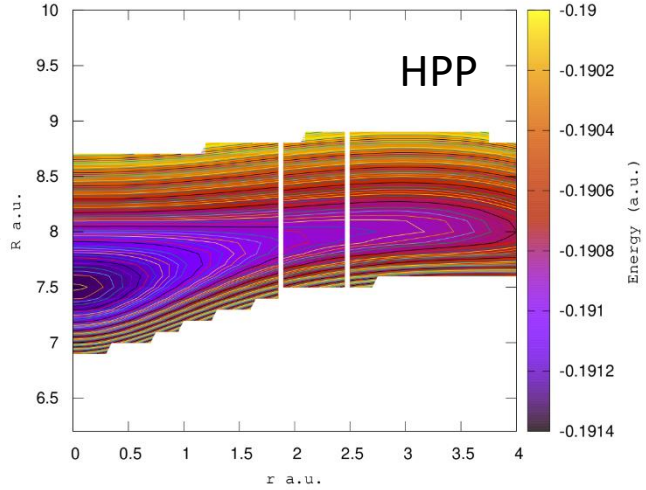
Why the difference?

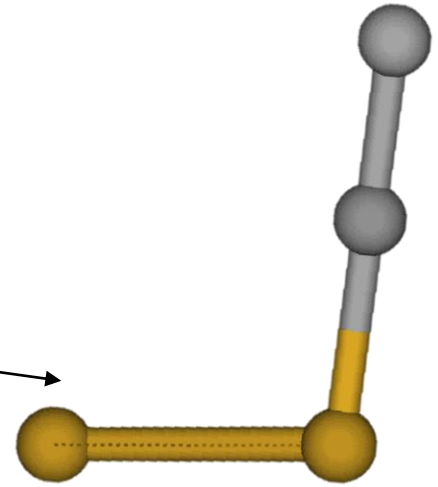




T-Shape





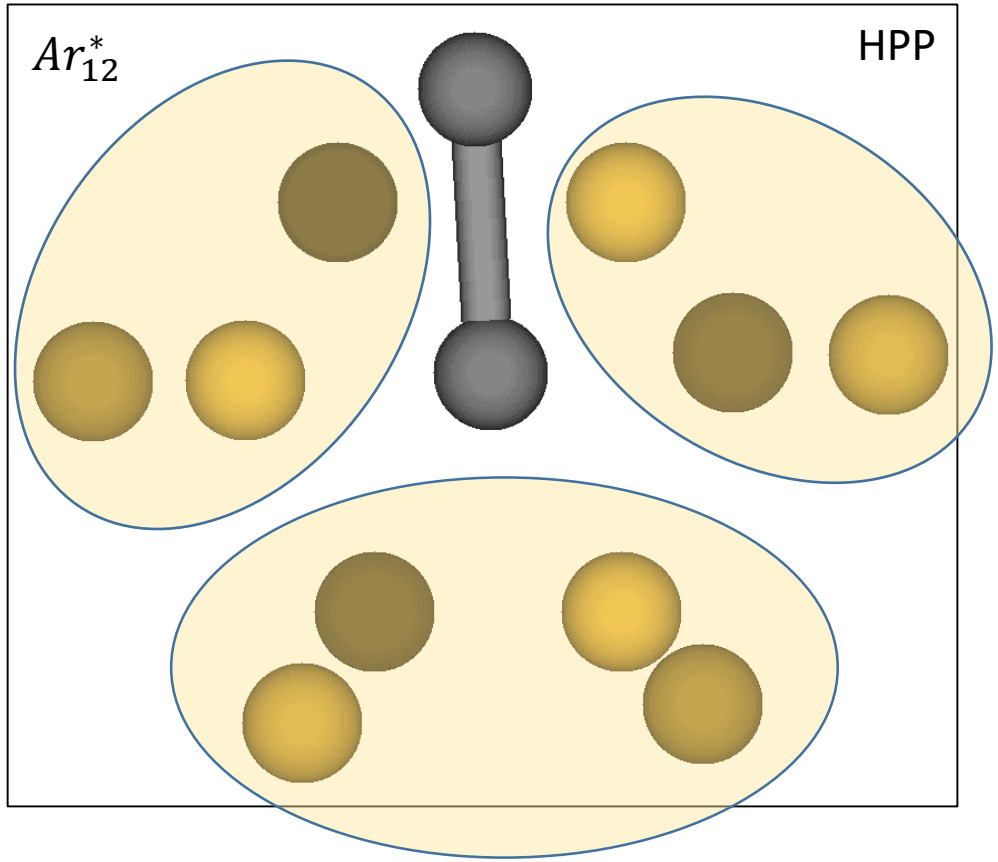


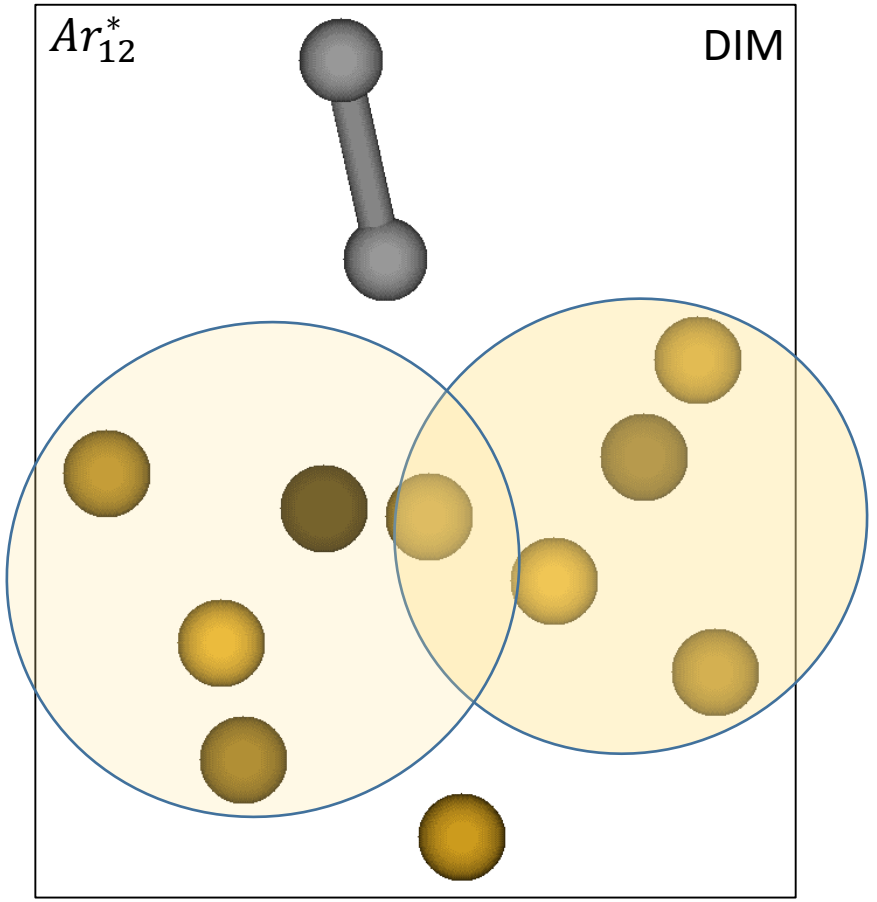
L-Shape

Isomer	Energy HPP (au)	Energy DIM	
Anti-Symmetric short	-0.19007293	-0.189112709	-9.6022E-04
triangular	-0.18754384	-0.187637343	9.3503E-05

Binding Energy
$Ar_2 \sim$ 4E-04

OTHER MINIMUM STRUCTURES





CONCLUSION

- The lowest trimer structure is not Linear-Symmetric as previously reported
- Diabatization is important to get true atomic and diatomic potential surface are essential for DIM method
- For the lowest energy structures of the excited neutral Argon cluster, the excitation is found to be protruding from the neutral cluster and is attached with one or more neutral atoms depending on HPP or DIM method
- In Hole-Particle Pseudopotential method, the excitation can accommodate more neutral atoms than DIM method due to absence of 4p orbitals
- To further improve the DIM we need to include higher (4p) states

CiMap

REFERENCES

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3. F. Y. Naumkin and D. J. Wales, “Rydberg excitations in rare gas clusters: Structure and electronic spectra of Ar*n (3 ≤ n ≤ 25),” *Mol. Phys.*, vol. 96, no. 9, pp. 1295–1304, 1999, doi: 10.1080/00268979909483074.
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5. P. Duplàa and F. Spiegelmann, “A pseudopotential hole-particle treatment of neutral rare gas excimer systems. II. the Rydberg states of the Ar2* dimer,” *J. Chem. Phys.*, vol. 105, no. 4, pp. 1500–1515, 1996, doi: 10.1063/1.472012.

THANK YOU FOR YOUR ATTENTION.

